CONTACT INFORMATION

JACKET DEMBY'S

Research Interests

- Robotics
- Machine Learning
- Deep Learning

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SUMMARY

First year Ph.D. in Electrical and Computer Engineering with growing interests in Robotics, Machine Learning and Deep Learning. Seeking research/industry experiences in Machine Learning and Deep Learning applied in Robotics applications.

EDUCATION

Master of Science Present GPA: 3.696/4	University of Missouri, Columbia, USA **Computer Engineering** Expected graduation: May 2020
Engineering Diploma Graduated July 2015 GPA: 15.06/20	Ucac-Icam Institute, Douala, Cameroon **Generalist Engineering** Top of promotion

RESEARCH EXPERIENCE

January 2020 – Present: Research Assistant (Vision Guided and Intelligent Robotics Lab – ViGIR Lab)

- **A study on the approximate solutions provided by data-driven and numerical inverse kinematics solvers for serial robotic manipulators.**

July 2019 – Present: Graduate Student

- **Object detection and pose estimation using CNN in embedded hardware for assistive technology:** I am involved on a research path related to object detection and pose estimation based on stereo vision and deep learning algorithms to be used on embedded devices for affordable assistive technology.

December 2018 – Present: Graduate Student

- **Inverse kinematics of serial robotics arms using soft-computing methods:** I am involved in looking at the inverse kinematics results' accuracies provided by two soft-computing methods: artificial neural network and fuzzy neural network.

August 2018 – December 2018: Graduate Student

- **Inverse kinematics of serial robotics arms using analytical and hybrid methods:** I was involved in looking at the effects of the unit choices for robot joints in the computation of the inverse kinematics using analytical and hybrid methods.

TEACHING EXPERIENCE

Spring 2020: **Teaching Assistant,** University of Missouri-Columbia, Columbia, Missouri, United-States

- **Class: Neural Networks**

PUBLICATIONS

1. **J. Demby's**, Y. Gao and G. N. DeSouza, "A Study on Solving the Inverse Kinematics of Serial Robots using Artificial Neural Network and Fuzzy Neural Network," *in 2019 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE),* New Orleans, LA, USA, 2019, pp. 1-6.
2. **J. Demby's**, Y. Gao, A. Shafiekhani, G.N. DeSouza, "Object Detection and Pose Estimation Using CNN in Embedded Hardware for Assistive Technology, " *in 2019 IEEE Symposium Series on Computational Intelligence (IEEE SSCI 2019)*, December 2019 (**accepted**)

Relevant Coursework

- Introduction to Mechatronic and Robotic Vision
- Probability and Stochastic Processes for Engineers
- Introduction to Machine Learning and Pattern Recognition
- Parallel Programming for High Performance Computing
- Introduction to Computational Intelligence
- Digital Image Processing
- Computer Vision
- Supervised Learning
- Neural Networks and Deep Learning
- Real-Time Embedded Systems

Some Coursework Projects involving Final Presentations

Class: Neural Networks (Spring 2019)
Project: Solving the inverse kinematics of a 4 DoF SCARA robot using extreme learning machines and radial basis function networks
Main objective, results and tools: Conduct a comparative study between extreme learning machines and radial basis function networks in solving the inverse kinematics of the chosen robotic arm. Lowest MSE (Mean-Squared Error) results were obtained with extreme learning machines. Matlab and Python were used.

Class: Introduction to Computational Intelligence (Fall 2018)
Project: Solving the inverse kinematics of a 6 DoF Kinova robot using artificial and fuzzy neural networks
Main objective, results and tools: Conduct a comparative study between artificial and fuzzy neural networks in solving the inverse kinematics of the chosen robotic arm. A trade-off was observed in terms of precision of the predicted joint configuration depending on the workspace. Matlab and Python were used.

Class: Computer Vision (Spring 2018)
Project: Hardware integration of a real time object detection CNN (Yolo V2)
Main objective, results and tools: Develop an android mobile application based on YOLO V2 to recognize and pronounce objects in real-time. Android, Darknet and a pretrained Yolo V2 model were used.

Class: Introduction to Mechatronic and Robotic Vision (Fall 2017)
Project: Investigating the control of a robotic arm with a real-time tracking of human hand motions
Main objective, results and tools: Track human hand motions using a Kinect sensor and control the robotic arm with the hand motions tracking information. C/C++ and ROS were used.

Class: Digital Image Processing (Fall 2017)
Project: Super Image Restoration (SIR)
Main objective, results and tools: Develop a Matlab application with a friendly Graphical User Interface (GUI) to restore old damaged images with cracks in only three click steps. Matlab, Python and Algorithmia Colorization API were used.

Achievements and Activities

May 2019 – present	Student Member of IEEE.
December 2017	Best 3rd final project in Digital Image Processing class (Fall 2017).
August 2017 – present	Member of the University of Missouri Fulbright Organization (**MUFO**).
July 2017 – present	Recipient of the **Fulbright Scholarship** in Congo Brazzaville to pursue a M.Sc. in Computer Engineering at the University of Missouri-Columbia.
January 2017	Participant for the Business and Entrepreneurship Track of the Young African Leaders Initiative (YALI Program) in Nairobi, Kenya.
April 2016	2nd prize ($13,000) of the 2015-2016 national Startup Challenge of Total E&P in Congo Brazzaville.
January 2015 – June 2015	Participant of the internal Icam exchange program to conduct an industrial mission on Computational Fluid Dynamics (CFD) in Lille, France.
June 2012 – July 2013	Recipient of the **Sogea-Satom scholarship** to study at Ucac-Icam Cameroon.
August 2010 – June 2012	Recipient of the **Total E&P scholarship** to study at Ucac-Icam Congo.

TECHNOLOGIES

Operating Systems: Linux, Windows, Mac
Programming languages: C/C++, Python, MATLAB, Android, Cuda, Lyx/Latex
Software and libraries:

- **Robotics**: ROS (Robot Operating System)
- **Machine learning**: TensorFlow, Keras
- **Computer vision**: OpenCV
- **Mechanical**: Solidworks, Ansys Fluent

WORK EXPERIENCE

Junior Project Engineer | December 2016 – July 2017
Total E&P Congo - Pointe-Noire, Republic of Congo

- Prepared, supervised and generated daily reports for construction activities of the Floating Production Unit (FPU) Likouf alongside contracted companies.
- Worked on Pressure Safety Valves (PSV) Recalibration Campaign, managed a database of the PSV joint stocks, and ensured team members compliance with on-site safety measures.

Maintenance Engineer | September 2015 – October 2016
Sogea-Satom - Brazzaville, Republic of Congo

- Monitored the computer-assisted supply chain and maintenance management software and produced monthly analysis reports.
- Developed and maintained Excel VBA (Visual Basic for Applications) programs for monthly stocks' management analysis, optimization and periodic inventories follow up.

Intern Maintenance Engineer by Alternance | January 2013 – December 2014
Sogea-Satom - Brazzaville, Republic of Congo

- Monitored the computer-assisted supply chain and maintenance management software.
- Continually improved methods and procedures for processes, stocks' management, documenting and workflow techniques.

Intern Maintenance Technician | January 2012 – March 2012
Total E&P Congo – Pointe-Noire, Republic of Congo

- Determine the causes of cascade regulations of three-phase separators of Katina treatment unit at Djeno Oil Terminal and propose improvement solutions to ameliorate their control/regulation cycles.
- Conduct a study on the appropriate type of sensors (ultrasonic sensors to detect water, gas and oil levels) to be replaced on those separators and suggest vendors.

Intern Operator | June 2011 – July 2011
Apave Congo – Pointe-Noire, Republic of Congo

- Certify welding structures (appliances and lifting gears) using radiography, dye penetrant and magnetic particles methods.

LANGUAGES AND INTERESTS

Languages

- French: native language
- English: fluent (speaking, reading, writing)

Interests

Sport: Walking, Running, Volleyball, Swimming
Board games: Scrabble, Ludo, Chess, Checkers

VOLUNTEER WORK

Intervarsity at University of Missouri-Columbia | Volunteer photographer and events' organizer

- Organize events (games, discoveries, English clubs, shopping, etc.) to help international students feel welcome at the university.
- Take pictures during the events and share them with all the members.

University of Missouri Fulbright Organization (MUFO) | Volunteer

- Work with the International Center to provide rides to new international sponsored students.
- Lead scavenger hunt activities to help international sponsored students discover and locate key facilities and resources on campus.

OVA (Œuvre de Vie et d'Amour) Association in Cameroon | Volunteer

- Organize activities on the International Day of the African Child and Christmas for street children and children with special needs.